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Ph.D. in Biology

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Education

Sun Yat-sen University, Guangzhou, Guangdong, China 2020 – 2026

Ph.D. in Biology

Kunming Medical University, Kunming, Yunnan, China 2017 – 2020

Master of Medicine in Medical Neurobiology

Zhejiang Chinese Medical University, Hangzhou, Zhejiang, China 2012 – 2017

Bachelor of Medicine in Clinical Medicine

Research Interests

- Profiles and mechanisms of neuronal differentiation
- Multi-omics library preparation and integrative analysis

Publications

† Equal contribution; * Corresponding author.

1. **Xu Y**, Yang J, Yuan W, Xu T, Shi P, Huang T, Ma Q, Xu J*. Single-cell long-read profiling of olfactory sensory neuron differentiation and diversity. *Communications Biology*. 2026.
2. Xu T†, **Xu Y**†, An Z, Li Y, Yang J, Zhang W, Xu J*. Methanol fixation and tagmentation of RNA/DNA hybrids directly enable single-cell transcriptome sequencing. *Frontiers in Genetics*. 2025.
3. Li X†, **Xu Y**†, Zhang W†*, Chen Z, Peng D, Ren W, Tang Z, Li H, Xu J*, Shu Y*. Immunoregulatory programs in anti-N-methyl-D-aspartate receptor encephalitis identified by single-cell multi-omics analysis. *Clinical and Translational Medicine*. 2025.
4. Zhang W†, Nie Y†, Xu T†, Li Y†, **Xu Y**, Chen X, Shi P, Liu F, Zhao H, Ma Q, Xu J*. Evolutionary process underlying receptor gene expansion and cellular divergence of olfactory sensory neurons in honeybees. *Molecular Biology and Evolution*. 2025.
5. Tang Z†, Zhang W†, Shi P†, Li S, Li X, Li Y, **Xu Y**, Shu Y, Hu Z, Xu J*. MitoSort: Robust demultiplexing of pooled single-cell genomic data using endogenous mitochondrial variants. *Genomics, Proteomics & Bioinformatics*. 2024.
6. Yang J, Shi P, Li Y, Zuo Y, Nie Y, Xu T, Peng D, An Z, Huang T, Zhang J, Zhang W, **Xu Y**, Tang Z, Li A, Xu J*. Regulatory mechanisms orchestrating cellular diversity of $Cd36^{+}$ olfactory sensory neurons revealed by scRNA-seq and scATAC-seq analysis. *Cell Reports*. 2024.
7. Liu B, **Xu Y**, Zhang W*. Transcriptome analysis of *Apis mellifera* antennae reveals molecular divergence underlying the division of labour in worker bees. *Insect Molecular Biology*. 2024.
8. Hu W, Xie Q, **Xu Y**, Tang X*, Zhao H*. Integrated bioinformatics analysis reveals function and regulatory network of miR-200b-3p in endometriosis. *BioMed Research International*. 2020.
9. Tang X, **Xu Y**, Lu L, Jiao Y, Liu J, Wang L, Zhao H*. Identification of key candidate genes and small molecule drugs in cervical cancer by bioinformatics strategy. *Cancer Management and Research*. 2018.

Software Development

- **CellMerge2**: An image acquisition, stitching, and analysis tool for immunofluorescence analysis
- **ATACFragQC**: A quality-control toolkit for ATAC-seq data
- **mrnaScan**: Read-level analysis of exon combinations and 3'-UTR usage in full-length mRNA-seq data